Course

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1.1.1. Calculate Momentum

09:32

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:
m is the mass of the object (in kilograms).
v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases +

calculate...

SUBMIT

```python
m=float(input())
v=float(input())
p=m*v
print(f"{p:.2f}kgm/s")
```

Terminal Test cases

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1.1.2. Conditional Calculation Based on the Number of Digits

07:59

Write a Python program that accepts an integer n as input. Depending on the number of digits in n.

Constraints:
$1 \le n \le 999$

Input Format:
- The input consists of a single integer n.

Output Format:
- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases +

Explorer

condition...

SUBMIT

```python
import math
n=int(input())
if(0<n<10):
 print(n*n)
elif(10<=n<100):
 print(f"{math.sqrt(n):.2f}")
elif(100<=n<1000):
	print(f"{n**(1/3):.2f}")
else:
	print("Invalid")
```

Debugger

Terminal Test cases

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1.1.3. Days between Two Dates

14:48

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format $YYYY - MM - DD$.
- The second line contains the second date in the format $YYYY - MM - DD$.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases +

Explorer

noOfDay...

SUBMIT

Debugger

```python
from datetime import datetime

date1=input().strip()
date2=input().strip()
d1=datetime.strptime(date1,"%Y-%m-%d")
d2=datetime.strptime(date2,"%Y-%m-%d")
difference=(d2-d1).days
print(difference)
```

Terminal Test cases

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1.1.4. Reverse a Number

06:42

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases +

Explorer

reverseN...

SUBMIT

```python
#write your code here...
num=int(input())
reverse=0
while num>0:
    digit=num%10
    reverse=reverse*10+digit
    num=num//10
print(reverse)
```

Debugger

Terminal Test cases

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1.1.5. Multiplication Table

18:24

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

```
<number> x <multiplier> = <result>
```

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases +

multiplica...

SUBMIT

```python
#write your code here...
num=int(input())
multiplier=1
while multiplier<=10:
    result=num*multiplier
    print(num,"x",multiplier,"=",result)
    multiplier+=1
```

Terminal Test cases

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1.2.1. Pass or Fail

14:37

Write a Python program that accepts the number of courses and the marks of a student in those courses.
The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n, the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

Explorer

passorFa...

SUBMIT

```python
#write your code here
#number of subjects
n=int(input())
#marks in each subject
marks=list(map(int,input().split()))
for m in marks:
    if m<40:
        print("Fail")
        break
else:
    average=sum(marks)/n
    print(f"Aggregate Percentage: {average:.2f}")

    if average>75:
        print('Grade: Distinction')
    elif average>=60:
        print("Grade: First Division")
    elif average>=50:
        print("Grade: Second Division")
    elif average>=40:
        print("Grade: Third Division")
```

Debugger

Terminal Test cases

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1.2.2. Fibonacci Series

06:47

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases +

Explorer

fibonacci....

SUBMIT

```python
def fibonacci(n):

    if n==1:
        return 0
    elif n==2:
        return 1
    else:
        return fibonacci(n-1) + fibonacci(n-2)


n = int(input())
for i in range(1, n + 1):
    print(fibonacci(i), end=" ")

```

Debugger

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1.2.3. Pattern - 1

01:23

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases +

Explorer

rightangl...

SUBMIT

```
#Type Content here...
n=int(input())
for i in range(1,n+1):
    for j in range(i):
        print("*",end=" ")
    print()
```

Terminal Test cases

Debugger

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1.2.4. Pattern - 2 01:27

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

Sample Test Cases +

numberP... SUBMIT

```python
# Type Content here...
n=int(input())
for i in range(1,n+1):
    for j in range (1,i+1):
        print(j,end=" ")
    print()
```

Terminal Test cases

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2.1.1. List operations

35:45

Write a Python program that implements a menu-driven interface for managing a list of integers. The program should have the following menu options:
1. Add
2. Remove
3. Display
4. Quit

The program should repeatedly prompt the user to enter a choice from the menu. Depending on the choice selected, the program should perform the following actions:

- **Add**: Prompts the user to enter an integer and add it to the integer list. If the input is not a valid integer, display "Invalid input".
- **Remove:** Prompts the user to enter an integer to remove from the list. If the integer is found in the list, remove it; otherwise, display "Element not found". If the list is empty, display "List is empty".
- **Display**: Displays the current list of integers. If the list is empty, display "List is empty".
- **Quit**: Exits the program.
- The program should handle invalid menu choices by displaying "Invalid choice".Ensure that the program continues to prompt the user until they choose to quit (option 4).

Sample Test Cases +

Explorer

listOps.py

SUBMIT

```python
nums=[]
while True:
    print("1. Add")
    print("2. Remove")
    print("3. Display")
    print("4. Quit")
    choice=input("Enter choice: ")
    if choice=="1":  # Add
        val=input("Integer: ")
        try:
            num=int(val)
            nums.append(num)
            print("List after adding:",nums)
        except:
            print("Invalid input")

    elif choice=="2":  # Remove
        if len(nums)==0:
            print("List is empty")
        else:
            val=input("Integer:")
            try:
                num=int(val)
                if num in nums:
                    nums.remove(num)
                    print("List after removing:",nums)
                else:
                    print("Element not found")
            except:
                print("Invalid input")

    elif choice=="3":  # Display
        if len(nums)==0:
            print("List is empty")
        else:
            print(nums)

    elif choice=="4":  # Quit
        break
    else:
        print("Invalid choice")
```

Terminal Test cases

< PREV RESET SUBMIT NEXT >

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2.1.2. Dictionary Operations

14:43

Write a Python program to perform insertion, update, deletion, and traversal operations on a dictionary. An initial dictionary containing 10 predefined records is already given in the program.

Operations to be Performed:

1. Insertion – Insert a new key-value pair into the dictionary using user input.
2. Update – Update the value of an existing key using user input.
3. Deletion – Delete a specified key from the dictionary using user input.
4. Traversal – Traverse the final dictionary and display all key-value pairs.

Input and Output Format:

1. Read an integer representing the key to be inserted and a string representing its value. Insert this new key-value pair into the dictionary and display the dictionary after insertion.
2. Read an integer representing the key to be updated and a string representing the new value. Update the value of the specified key (only if it exists) and display the dictionary after the update.
3. Read an integer representing the key to be deleted. Delete the specified key from the dictionary (only if it exists) and display the dictionary after deletion.
4. Finally, traverse the dictionary and display all key-value pairs.

The program should also display the original dictionary before performing any operations. Each output should be printed with appropriate messages.

Note:

- All operations must be performed using dictionary methods.
- Perform deletion only if possible; leave the dictionary unchanged.
- Refer to the visible test cases for better understanding and strictly match with the input/outputs.

Sample Test Cases

DictOper...

SUBMIT

```
    3: "Kiran",
    4: "Neha",
    5: "Arjun",
    6: "Pooja",
    7: "Rahul",
    8: "Sneha",
    9: "Vikram",
    10: "Anjali"
}

# write your code here...
print("Original Dictionary:",student)
key=int(input())
value=input()
student[key]=value
print("After Insertion:", student)
key=int(input())
value=input()
if key in student:
    student[key]=value
print("After Update:", student)
key=int(input())
if key in student:
    del student[key]
print("After Deletion:", student)
print("Traversing Dictionary:")
for k,v in student.items():
    print(k, ":", v)
```

Terminal Test cases

PREV RESET SUBMIT NEXT

ENG 09:06 PM 21-04-2026

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2.2.1. Linear search Technique

03:08

Write a program to check whether the given element is present or not in the array of elements using linear search.

Input format:

- The first line of input contains the array of integers which are separated by space
- The last line of input contains the key element to be searched

Output format:

- If the element is found, print the index.
- If the element is not found, print **Not found**.

Sample Test Case:

Input:

1 2 3 4 3 5 6

3

Output:

2

Sample Test Cases +

CTP1709... SUBMIT

```python
arr=list(map(int,input().split()))
key=int(input())
found= False
for i in range(len(arr)):
    if arr[i]==key:
        print(i)
        found=True
        break
if found == False:
    print("Not found")
```

Terminal Test cases

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2.2.2. Captain of the Team

04:38

You are provided with the heights of 11 cricket players (in centimeters). Your task is to identify the tallest player, who will be selected as the captain of the team.

Input Format:
The first line of input will contain 11 integers, each representing the height of a player (in centimeters), each separated by a space.

Output Format
The output should be the height (in centimeters) of the tallest player.

Sample Test Cases +

Explorer

captainof...

SUBMIT

```python
arr=list(map(int,input().split()))
tallest=arr[0]
for h in arr:
    if h>tallest:
        tallest=h
print(tallest)
```

Debuger

Terminal Test cases

< PREV RESET SUBMIT NEXT >